

CAPE PHILLIPS, Antarctica

WMO: 896610

Lat: 73.518S

Lon: 169.749E

Elev: 1867

StdP: 13.73

Time Zone: 12.00 (NZL)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -29.6	(c) -27.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 84.3	(k) -12.6	(l) 78.5	(m) -6.5	(n) 14.2	(o) 240	(p) 1.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 5.0	(c) 32.1	(d) N/A	(e) 29.0	(f) N/A	(g) 26.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 8.1	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
79.0	68.0	58.9	-33.0	37.6	1.8	3.4	-34.2	40.0	-35.3	42.0	-36.3	44.0	-37.6	46.5	
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	22.2	14.9	4.2	-5.4	-6.3	-10.0	-12.2	-9.3	-7.9	-2.3	11.3	20.9		
	DBStd	14.09	3.22	4.26	6.25	8.07	9.36	9.04	8.82	9.17	8.64	6.17	6.85	4.15		
	HDD50	17657	861	984	1421	1661	1744	1800	1928	1839	1738	1620	1160	901		
	HDD65	23132	1326	1404	1886	2111	2209	2250	2393	2304	2188	2085	1610	1366		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	19.4	17.9	18.6	19.4	17.7	19.7	21.8	20.9	22.1	21.3	22.8	17.6	13.4		
	PrecAvg	14.3	1.7	1.9	1.5	1.0	1.1	1.1	0.9	1.1	1.1	1.1	1.0	1.1		
(15) Precipitation	PrecMax	19.6	3.2	4.6	2.8	2.1	2.3	2.1	2.1	2.4	2.3	2.2	2.3	2.1		
	PrecMin	8.7	0.5	0.7	0.8	0.3	0.2	0.2	0.3	0.3	0.0	0.4	0.3	0.3		
	PrecStd	2.4	0.7	0.8	0.5	0.5	0.6	0.4	0.4	0.6	0.6	0.5	0.6	0.5		
	PrecStd	2.4	0.7	0.8	0.5	0.5	0.6	0.4	0.4	0.6	0.6	0.5	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	27.8	17.5	19.1	20.5	14.2	11.2	21.5	16.3	18.3	31.2	37.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	31.4	23.8	15.7	12.6	13.5	8.9	7.2	12.1	11.3	13.0	27.2	32.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	29.0	22.0	14.1	9.1	9.1	6.0	2.7	6.6	8.1	8.8	23.0	29.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	26.9	20.5	12.3	4.6	5.8	2.9	2.0	2.8	4.4	6.2	20.2	26.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	4.9	5.4	7.3	8.7	9.0	8.6	10.2	9.1	6.7	6.9	6.0		
		MCDBR	9.2	7.1	7.5	12.3	14.5	12.1	11.6	17.5	15.3	10.6	9.5	10.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.238	0.197	0.110	N/A	N/A	N/A	0.108	0.174	0.197	0.218	0.237		
	taud		2.427	2.422	2.333	1.922	N/A	N/A	N/A	1.901	2.250	2.334	2.343	2.366		
	Ebn at Noon		314	293	266	173	0	0	0	167	272	309	323	323		
	Edh at Noon		28	23	17	6	0	0	0	6	17	25	30	31		
(44) All-Sky Solar Radiation	RadAvg		1968	1033	497	116	0	0	0	40	442	1277	2274	2651		
	RadStd		220	140	77	8	0	0	0	3	24	49	87	169		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page